

RTL Design Sherpa

Pre-Synth Timing Prediction A White Paper on Designing-for-Close Before Synthesis Ever Starts

June 12, 2026

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1 Readme

Generated: 2026-10-02

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2 A Portable, Macro-Optional ASIC Timing Characterization Harness

> A practitioner's guide to `char_top` – a benchmarking harness that measures combinational delay across ASIC standard-cell libraries by forcing the synthesis tool to build the structures we actually want it to build.

Version: 3.1 (ASAP7 worked example landed) **Last Updated:** 2026-06-12 **Status:** RTL + verification + ASIC synthesis flows in place; ASAP7 RVT characterization run + budget worksheet live under `work/`. (45 tests passing; Yosys + OpenSTA + Synopsys DC + Cadence Genus drivers; 99-row ASAP7 sweep at TT/FF/SS clean.)

> An FPGA-side companion – with the same workflow and an analogous spreadsheet – is published as `README_FPGA.md` in this same directory. This document is ASIC-only.

2.1 Abstract

The sole purpose of this methodology is to allow the designer to predict and close timing **before synthesis ever starts**. This flow lets the designer characterize various structures at various technology nodes and frequencies so that, by the time the production design enters synthesis, each combinational shape has a known per-node, per-frequency budget – and the answer to "will this pipeline stage close?" is already on a table. The same characterization run also yields an **accurate per-structure gate count**: because the harness pins every internal net through `dont_touch` and emits a per-FUB cell-count + area summary from both Yosys (`build/$TOP.stat.txt`) and the commercial flows (`report_area`), the designer gets a real, technology-mapped gate-count estimate for any combinational shape long before the production netlist exists.

This methodology is **explicitly proactive**. It is not a debugger you reach for after a synthesis run reveals a violation; it is a design-time budget the designer (or uArch) consults *before writing the first `always_ff`*. The whole point is to flip the loop – characterize once per node, then build with a known-good budget – so synthesis becomes a confirmation step rather than the first time anybody finds out the pipeline doesn't close. Concretely, the methodology exists **to prevent, as much as possible, the designer being surprised when synthesis happens**: every shape on the critical path has already been measured at this node and frequency, every gate count has already been totalled, and the answer is on the worksheet long before the synthesis tool ever weighs in.

`char_top` is a synthesizable harness that wraps nine **Functional Unit Blocks (FUBs)** – each a different shape of combinational logic – between registered endpoints driven by a shared 32-bit Galois LFSR. By driving every flop input with a maximal-length pseudo-random sequence and

routing every flop output to a top-level port, the design forecloses every optimisation a modern synthesis tool would otherwise use to make the benchmark disappear (constant propagation, dead-code elimination, common-subexpression elimination, retiming-into-pads, and so on). Within that frozen scaffold, each FUB exposes a specific combinational pattern (NAND tree, ripple-carry adder, mux tree, structural multiplier, …) so that the slack reported for its path group is a clean, comparable measure of that pattern's depth in the target technology.

There is ONE source tree, `rtl/`, macro-driven, and it serves both FPGA and ASIC. It ships with an open-source Yosys + OpenSTA flow (`rtl/syn/`) plus reference scripts for Synopsys Design Compiler and Cadence Genus.

There used to be a second tree, `rtl/asic_only/` – a 37-file duplicate with `ALWAYS_FF_RST` expanded and `fifo_mem_t` hand-substituted for `int`. It was deleted on 2026-09-07 because it bought nothing. Synthesised through Yosys + slang against ASAP7, `char_top` from the two trees produced identical mapped netlists: 4753 cells, 4598 wires, all 50 cell-type counts matching. The macros exist to serve ASIC and FPGA both; a second copy of every module to spell reset differently is duplication, not portability.

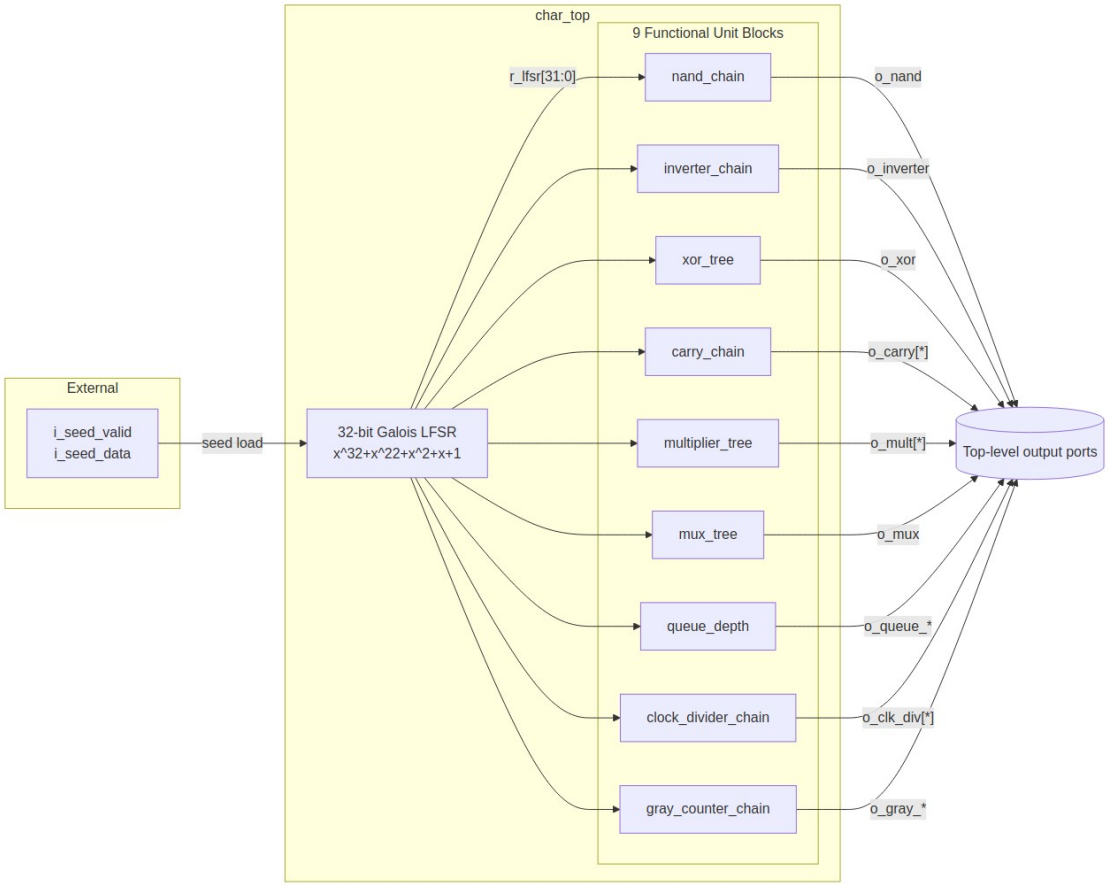
2.2 1. Why this harness, and not just RTL we already have?

A synthesis tool is, by design, free to delete logic whose output is unobserved or whose input is constant. Without explicit countermeasures any benchmark "design" collapses to a few wires before it ever reaches mapping. This harness solves four problems at once:

1. it gives every standard combinational pattern (tree, chain, multiplier, mux, memory, counter) a **registered envelope** so STA reports the pattern's own delay, not an I/O delay;
2. it drives every flop input from a **shared maximal-length LFSR** so the tool cannot prove any input constant;
3. it routes every flop output to a **top-level port** so the tool cannot prune any output;
4. it carries `(* dont_touch = "true" *)` on every chain net as a belt-and-suspenders defence against aggressive remapping.

The result is a single source that runs through any flow you point at it and lets you measure – not estimate – how a 12-level XOR tree, or a 64-bit ripple-carry adder, or a 16x16 Dadda multiplier behaves in the target technology.

2.3 2. High-level architecture

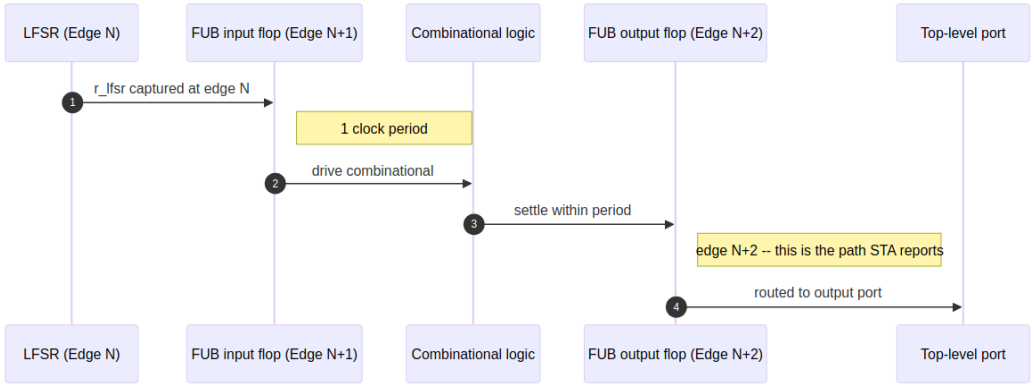


Architecture overview

Each FUB is gated by an `EN_*` parameter. When disabled (`EN_*=0`) the corresponding generate block emits a tie-to-zero and consumes zero logic – this is how targeted "just the multiplier" or "everything but the FIFO" sweeps work.

2.4 3. The 3-edge pipeline

The harness deliberately registers both endpoints of every FUB's combinational logic so STA reports slack for the register-to-register path itself, not the I/O path. Because the LFSR is the source for every input, the design has a fixed three-edge pipeline:

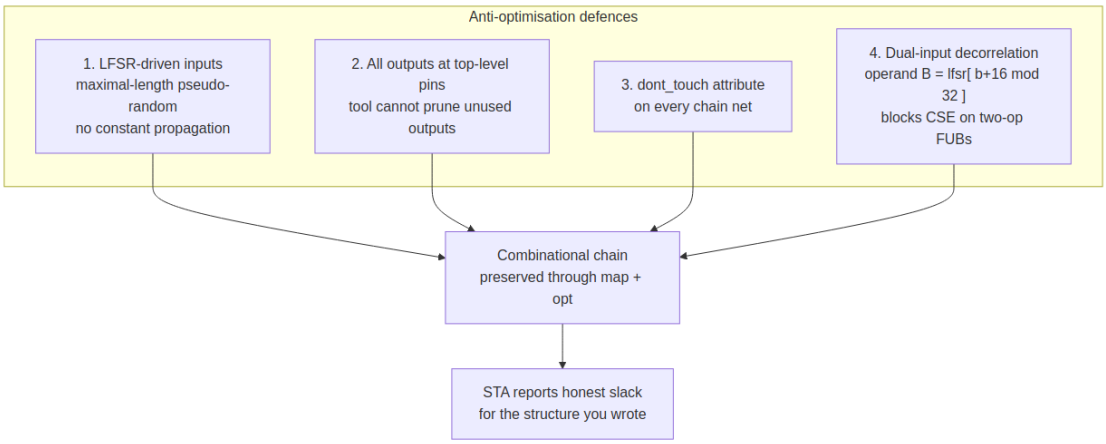


3-edge pipeline

A char_top-level testbench that compares an expected value against an observed output sampled after rising edge C reads `lfsr_idx = C - 2` from the reference model (`dv/tbclasses/timing_char_tb.py:lfsr_step / lfsr_sequence`).

The LFSR in `char_top` is a **left-shift** Galois LFSR (polynomial `0x0040_0007`). This is architecturally different from the right-shift tap-based Galois LFSR in `val/common/test_shifter_lfsr_galois.py` – do **not** reuse that model.

2.5 4. Anti-optimisation strategy



Anti-optimisation defences

Defence	Threat it blocks
LFSR-driven inputs	Constant propagation, dead-input elimination
Outputs at the pads	Dead-output elimination, sweep

Defence	Threat it blocks
`dont_touch` attribute	Aggressive remapping that erases internal nets
Dual-input decorrelation (`r_lfsr[(b+16) % 32]`)	CSE on `a OP a`

All four work together. Remove any one and a sufficiently aggressive synthesis tool will find a shortcut; with all four in place the gates you wrote are the gates that land in the netlist.

2.6 5. The FUB catalog

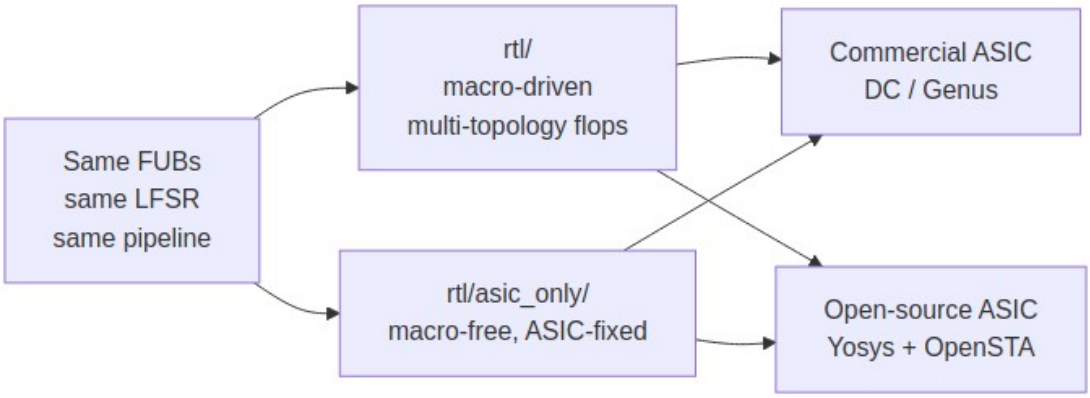
FUB	File	Key parameters	Critical path	What it measures
`nand_chain`	`rtl/fub/nand_chain.sv`	`LEVELS`, `NUM_FLOPS`	LEVELS NAND-gate delays	Raw 2-input NAND propagation through the target technology
`inverter_chain`	`rtl/fub/inverter_chain.sv`	`INVERTER_COUNT`	Linear inverter chain	Best-case per-gate delay (no fan-in overhead)
`xor_tree`	`rtl/fub/xor_tree.sv`	`LEVELS`, `NUM_FLOPS`	LEVELS XOR delays	XOR cell vs. NAND cell / LUT packing
`carry_chain`	`rtl/fub/carry_chain.sv`	`WIDTH`	WIDTH carry-propagation delays	Dedicated carry-chain (CARRY4 / CARRY8 / ALM-carry / stdcell ripple)
`multiplier_tree`	`rtl/fub/multiplier_tree.sv`	`WIDTH`, `MULT_TYPE`	Reduction-tree depth	Inferred * (DSP) vs. structural

FUB	File	Key parameters	Critical path	What it measures
				Dadda / Wallace / CSA
mux_tree	rtl/fub/mux_tree.sv	LEVELS, NUM_FLOPS	LEVELS 2:1-mux delays	Mux-cell delay vs. logic-cell delay
queue_depth	rtl/fub/queue_depth.sv	DATA_WIDTH, DEPTH	FIFO read path + output flop	Memory inference thresholds (LUTRAM / BRAM / URAM / SRAM compiler)
clock_divider_chain	rtl/fub/clock_divider_chain.sv	STAGES, CW, PICKOFF	Counter increment + pickoff mux	Counter-based clock generation
gray_counter_chain	rtl/fub/gray_counter_chain.sv	WIDTH	Binary carry + XOR tree	Binary-to-Gray conversion delay

Per-FUB rationale, recommended sweep ranges, and result interpretation live in `rtl/syn/SYNTHESIS_GUIDE.md`.

2.7 6. The two source trees

The component ships two source-equivalent trees so that the same nine FUBs can be characterised with the right amount of preprocessor indirection for the job.



Two source trees

2.7.1 6.1 `rtl/` – the macro-driven default

The default tree includes `reset_defs.svh` and `fifo_defs.svh` so a single source set covers every flop topology a target ASIC library might expose:

Switch	Effect
\RESET_ACTIVE_HIGH| Flip reset polarity | |MEM_STYLE = FIFO_{AUTO,SRL,BRAM}| Drive the inferred memory style ofgaxi_fifo_sync| | ~~`USE_ASYNC_RESET`~~	**No longer a switch.** Reset is asynchronous on assertion in every build; the define parses and does nothing

Use this tree when the cell library you're characterising has both active-high and active-low reset cells and you want a single source to sweep both.

`USE_ASYNC_RESET` was removed as a switch repo-wide on 2026-09-07: it defaulted to SYNCHRONOUS while `make lint` set the define, so lint and the shipped bitstream disagreed about what the design was. Making it unconditional fixed that, at the cost of this component's ability to characterise sync-reset cells from the shared macro – see section 12.

2.7.2 7.1 Open-source flow (Yosys + OpenSTA)

```
cd projects/asic-trials/timing_characterization/rtl/syn

# Minimal invocation -- 500 MHz, every FUB enabled
make report LIB_PATH=/path/to/stdcell.lib

# 800 MHz, no multiplier (isolates the rest of the harness)
make report LIB_PATH=/path/to/stdcell.lib \
            TARGET_FREQ_MHZ=800 EN_MULTIPLIER=0
```

```
# Just rerun STA against the existing netlist
make sta LIB_PATH=/path/to/stdcell.lib
```

Outputs land under `syn/build/`:

File	Contents
`build/$TOP.netlist.v`	Post-mapped gate-level netlist
`build/$TOP.json`	Yosys JSON (for graph tools, formal, ABC reruns)
`build/$TOP.stat.txt`	Cell-count + area summary per Liberty
`build/sta.log`	Full OpenSTA log incl. WNS per group
`build/timing/GRP_*.rpt`	Per-FUB worst-path report
`build/area.rpt`	Hierarchical area
`build/yosys.log`	Synthesis trace for diffing across runs

2.7.3 7.2 Commercial reference flows

```
dc_shell -f rtl/syn/dc_synth.tcl       # Synopsys DC
genus    -files rtl/syn/genus_synth.tcl # Cadence Genus
```

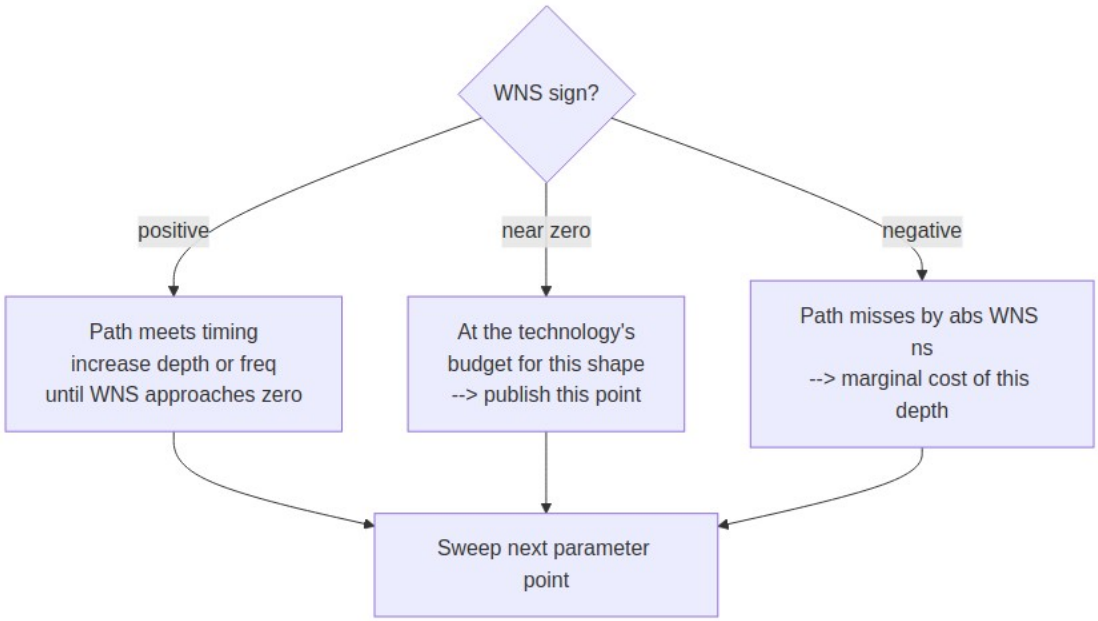
Both reference scripts source the same `char_top.sdc` and declare the same per-FUB path groups, so reports are directly comparable to the open-source flow's output.

2.8 8. Reading the reports

The interesting line in the OpenSTA log (or the DC / Genus equivalent) is the per-group worst-negative-slack:

```
Path Group: GRP_NAND       WNS =  0.1234 ns
Path Group: GRP_INVERTER   WNS =  0.4567 ns
...
```

Positive WNS = the FUB meets the target frequency with that much slack to spare. Negative WNS = the FUB misses by that much; the absolute value is the additional period the technology needs at this depth.



WNS-sign decision tree

Sweep protocol:

1. Fix `TARGET_FREQ_MHZ`.
2. For each FUB, sweep its depth / width parameter until WNS crosses zero.
3. Record the parameter value at WNS = 0 – that's the technology's max depth for that pattern at the chosen frequency.
4. Repeat across libraries / devices for an apples-to-apples cross-target view.

2.8.1 8.1 Comparing technologies

Run the same sweep against two libraries:

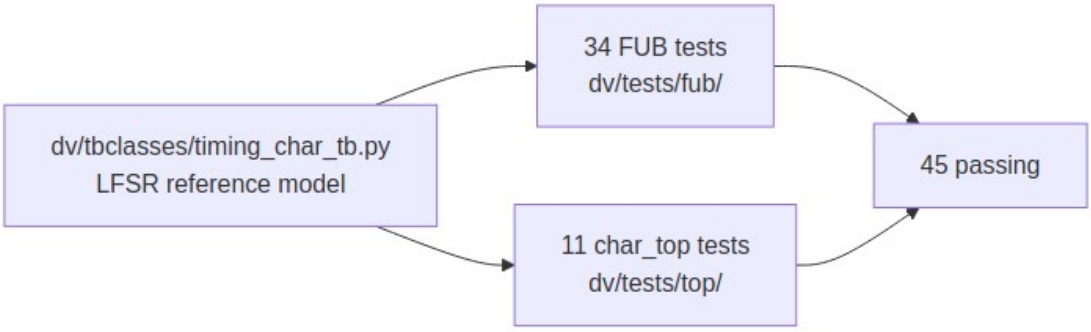
```
make report LIB_PATH=lib_A.lib TARGET_FREQ_MHZ=500
mv build/timing build/timing.A

make clean
make report LIB_PATH=lib_B.lib TARGET_FREQ_MHZ=500
mv build/timing build/timing.B

diff -u build/timing.A/GRP_CARRY.rpt build/timing.B/GRP_CARRY.rpt
```

The path structures are identical; only the cell delays differ. Any WNS delta therefore comes purely from the technology, not from a synthesizer mood swing or a netlist topology change.

2.9 9. Verification



Verification topology

Suite	Location	Tests	Status
Per-FUB sweeps	`dv/tests/fub/`	34	All passing
`char_top` enable-mix configurations	`dv/tests/top/`	11	All passing

```
PYTHONPATH=bin:$PYTHONPATH pytest \
    projects/asic-trials/timing_characterization/dv/tests/ -v
```

The TB class lives in `dv/tbclasses/timing_char_tb.py` and implements the mandatory three methods (`setup_clocks_and_reset`, `assert_reset`, `deassert_reset`); the `lfsr_step` / `lfsr_sequence` reference functions match the left-shift LFSR baked into `char_top`.

2.10 10. Directory layout

```
projects/asic-trials/timing_characterization/
├── README.md                       <-- this file
├── CLAUDE.md                       AI assistance guide
├── PRD.md                          Product Requirements
│                                    (task tracking moved to the vault,
2026-09-25)
├── assets/                         diagram sources + rendered PNGs
│   ├── Makefile                    `make` to re-render every *.mmd ->
*.png
│   ├── puppeteer.json              mmdc no-sandbox config
│   ├── *.mmd                       diagram source (mermaid)
│   └── *.png                       diagram renders (embedded above)
├── docs/                           Characterization results (future)
├── dv/
│   ├── tbclasses/
```

```
│   │   └── timing_char_tb.py    TB base + LFSR reference model
│   └── tests/
│       ├── fub/                 9 FUB tests (34 parameter points)
│       └── top/                 char_top tests (11 enable-mixes)
├── work/                        <-- the worked example artifacts
│   ├── asap7_characterization.xlsx  3-sheet methodology workbook
│   ├── timing_char_sweep.py     9-primitive 2-point probe driver
│   ├── timing_char_data.csv     54-row primitive characterization
data
│   ├── abc_sweep.py             11-row STREAM library sweep driver
│   ├── timings.csv              99-row per-block ASAP7 sweep output
│   ├── build_characterization_xlsx.py  CSV -> xlsx generator
│   └── merge_lib_corner.py      ASAP7 RVT Liberty merger (TT/FF/SS)
└── rtl/
    ├── common/                  shared primitives (counters, math,
gaxi_fifo_sync)
    ├── fub/                     9 FUBs
    ├── filelists/char_top.f     flat source list for the multi-flow
tree
    ├── top/                     char_top.sv + nand_chain_top.sv
    └── syn/                     ASIC + FPGA flow (the multi-flow SDC;
see also fpga/quartus/ for the Quartus sweep)
        ├── char_top.sdc         multi-flow constraints
(asic/vivado/quartus)
        ├── SYNTHESIS_GUIDE.md   per-FUB synthesis recipes
        ├── Makefile             Yosys + OpenSTA driver
        ├── yosys_synth.ys       reads via slang (typedef'd params +
`include)
        ├── sta_run.tcl
        ├── dc_synth.tcl         Synopsys DC reference
        └── genus_synth.tcl      Cadence Genus reference
```

2.10.1 10.1 Regenerating the diagrams

The seven figures in this document live as mermaid source in `assets/` and are rendered to PNG by `mmdc` (`@mermaid-js/mermaid-cli`). Re-render after editing the source:

```
cd projects/asic-trials/timing_characterization/assets
make            # render every *.mmd -> *.png
make clean      # delete the PNGs
```

`assets/puppeteer.json` passes `--no-sandbox` so headless Chrome runs inside sandboxed environments and CI runners.

2.11 11. Limitations and intentional non-features

- **No memory compiler integration.** `queue_depth` is mapped by the tool to whatever the Liberty file exposes; an SRAM-compiler wrapper is out of scope.
- **No SDF back-annotation.** STA against the synth Liberty is wire-load estimated, not parasitic-aware. Round-trip P&R + extraction is out of scope of a *characterization* harness.
- **No power numbers from the open-source flow.** Add a switching activity file and a power-aware Liberty to extend the OpenSTA invocation with `report_power`.
- **Async-reset only; sync-reset characterization is currently unavailable.** The harness models the worst case (async reset path through every flop) so the cell library's reset recovery / removal delay is reflected in the report. Sync-reset numbers used to come from this tree with `USE_ASYNC_RESET` left undefined; that switch became a no-op on 2026-09-07 when reset was made unconditionally asynchronous repo-wide. Recovering the capability means giving this component its own characterization-local reset header rather than sharing `rtl/amba/includes/reset_defs.svh` – deliberately, and named so it is obviously a measurement knob and not a copy of the design's header. Tracked as CHAR-001.

2.12 13. From characterization to design: the ASAP7 worked example

The methodology's design-time artifact is a single workbook with three linked sheets: `work/asap7_characterization.xlsx`. It's built bottom-up against ASAP7 RVT (the same flow described in section 7) and lets a designer answer "will this pipeline close?" by changing one cell – before any RTL exists for the production block.

The three sheets stack:

1. **`characterization`** – the technology data. Per-gate-type delay (NAND, XOR, MUX, CARRY, MULT, …) at TT / FF / SS, plus Tflop, wire delay per metal layer, and the **Design clocks table** that names the clocks the design uses.
2. **`building blocks`** – a SWAG estimator for every reusable RTL block (skid buffer, gaxi_fifo_sync, arbiter_round_robin, axi4_master_rd/wr, SRAMs).

Each row's data->D / flop->out / fmax / slack are formulas that look up the planning corner and freq from the characterization sheet.

3. **`stream`** – a worked example: every port of every STREAM FUB walked by hand, one row per (FUB, port) pair. The slack cell on each row is corner- and freq-aware via VLOOKUP into the clocks table.

Editing the clocks table on sheet 1 propagates to slack columns on sheets 2 and 3. Negative slack cells render **bold red** so failing paths jump off the page during the sweep.

2.12.1 13.1 Sheet 1: `characterization` – the technology data

The top of the sheet is a sweep grid: TT / FF / SS x five frequencies (currently 2.0 / 2.25 / 2.5 / 2.75 / 3.0 GHz). Each cell shows what's achievable in that corner+freq combination. The two driving rows are:

- **Per-level gate delay** (ps, freq-independent) – the slope from a two-point synth probe per primitive at that corner. This is what gets pulled by the downstream sheets.
- **Max combinational levels per cycle** – the budget: `(period - Tflop) / per_level_delay`. Tells you how deep a chain of that gate type can be before it busts the period.

TT corner, freq sweep	NAND	XOR	MUX	CARRY	MULT (proxy)	GRAY_CTR	QUEUE
per-level delay (ps)	7.25	26.8	16.3	30.8	**30.0**	58.4	11.6
max levels @ 2.0 GHz	66	18	29	15	**16**	8	41
max levels @ 3.0 GHz	43	12	19	10	**10**	5	27

The **MULT** row is highlighted yellow on the sheet – it doubles as the recommended proxy for mixed-bag combinational logic (a real mult tree contains AND, XOR, full-adder, and carry cells, so its per-level delay blends realistically).

Below the per-corner data lives the **Design clocks table** – one configurable row per clock the design uses (up to six slots):

clock name	freq (MHz)	target corner	period (ps)	input 60%	output 40%	wire/hop
clk_main (5 empty slots)	**1000**	**TT**	=1e6/B	=D*0.6	=D*0.4	100

Yellow cells are editable; the rest derive. This table is what sheets 2 and 3 read – their `slack` columns are wired here via VLOOKUP.

Last two sections on the sheet: **Wire delay per mm by metal layer** (M2/M3 through M8/M9 – 120 to 22 ps/mm TT), and notes.

2.12.2 13.2 Sheet 2: `building blocks` – per-block SWAGs

One row per reusable RTL block, in two sections. The **SRAMs** section holds the storage primitives:

block	params	flops	NAND_eq	data->D (TT, ps)	flop->out	slack @ clk_main
`SRAM (raw macro)`	W=512, D=512	0	~2110	7	7	(depends on clock)
`gaxi_fifo_sync (REG=1, SRAM-backed)`	W=512, D=512	28	2188	7	7	…

`flops` on the SRAM rows is the **control-only** count – storage bits (`W * D`) live in the BRAM/SRAM macro, not in std cells. A reference block below the table shows `storage = W*D` in bits and KB.

The **Building Blocks** section follows with the rest:

block	base params	flops	NAND_eq	in mux/nand	out mux/nand	slack @ clk_main
`gaxi_skid_buffer`	W=1, D=2	6	18	1 / 1	**0 / 0**	(computed)
`gaxi_fifo_sync (REG=0)`	W=1, D=2	6	40	1 / 2	log2(D) / 0	…

block	base params	flops	NAND_eq	in mux/nand	out mux/nand	slack @ clk_main
arbiter_round_robin	N=2	4	16	0 / 1	log2(N) / 1	…
axi4_master_rd	AW=DW=32, IW=4	212	848	1 / 1	1 / 1	…
axi4_master_wr	similar	222	888	1 / 1	1 / 1	…

Each row's `clock` column (yellow) defaults to `clk_main` – change it to point a block at a different clock entry on sheet 1. `data->D` covers the input-side combinational delay; `flop->out` covers the output-side; `delay_in` defaults to 30 percent of the target period (STA input arrival); `slack = period - delay_out - 40 percent of period - wire_per_hop`.

Skid buffers have `out_mux = out_nand = 0` because the head flop's Q drives the output directly – there's no internal output muxing. The other blocks have non-zero output-side levels (the read mux tree on FIFOs, the priority encoder on the arbiter, the skid output select on the AXI masters).

2.12.3 13.3 Sheet 3: `stream` – per-FUB port trace

This is the worked example proper: 11 STREAM FUBs, 144 rows total, one row per `(FUB, port)` pair. Walking `rtl/fub/stream_latency_bridge.sv` by hand gave the first ten rows; the other ten FUBs were filled in the same way.

Each row has:

col	content
A	FUB name
B	port name
C	clock (defaults `clk_main`; edit to point at a different clock)
D	direction (in / out / local)
E	building block crossed (if the port enters / leaves a BB)
F	MUX lv on this row's path
G	NAND lv on this row's path
H	source/sink flop or BB port the path terminates at
I	target period (VLOOKUP from clocks

col	content
	table)
J	target corner (VLOOKUP)
K	local delay (corner-aware: F * per_MUX(corner) + G * per_NAND(corner))
L	delay_in (30 percent of target period by default; override with `=M_prev` to chain)
M	delay_out = delay_in + local delay
O	**slack = period - delay_out - output_40% - wire/hop**
P	places it goes (which downstream FUB / CSR / system bus consumes this)

A `local` direction marks an internal flop hop (`r_state` -> `r_next` inside one FUB). 39 of the 144 rows are local-flop hops; 26 cross a building block boundary; the rest are external ports.

The “places it goes” column carries the partition story. For example:

FUB	port	dir	terminator	places it goes
`stream_latency_bridge`	`m_valid`	out	`u_skid_buffer.rd_valid (Q)`	to `sram_controller_unit.axi_wr_sram_valid` -> `axi_write_engine`
`scheduler`	`axi_rd_req_valid`	out	`r_axi_rd_req (Q)`	to `axi_read_engine` (per-channel request)
`descriptor_engine`	`descriptor_valid`	out	`i_descriptor_fifo.rd_valid (Q)`	to `scheduler` (descriptor dispatch)

2.12.4 13.4 The workflow: planning at the spreadsheet

The whole point of the three-sheet stack is to make pipeline planning a spreadsheet exercise:

1. Open the workbook, go to `characterization`, find the Design clocks table, set `clk_main` freq + corner to your target.
2. Flip to `building blocks` – which BB rows light up red in the slack column? Those are the blocks you need to either pipeline harder or stop relying on at this freq.
3. Flip to `stream` – which FUB ports are red? Those tell you which inter-FUB hops won't close. The "places it goes" column maps a red row back to a concrete RTL location.
4. Iterate: lower the freq until everything's green (the design's actual ceiling); switch to FF corner to model best-case silicon; bump wire/hop to model long routes.

Three knobs to play with:

- **`clk_main` freq** (B29) – walk the period; watch slack flip.
- **`clk_main` target corner** (C29) – TT for sign-off, FF for best-case silicon, SS for worst-case.
- **`clk_main` wire/hop** (G29) – default 100 ps; bump to 200-300 for long inter-FUB routes, drop to 0 for paths that stay inside one FUB.

Add more clocks (clk_apb, clk_cdc, …) in slots 30-34 and re-point specific rows on sheets 2 / 3 by editing their `clock` cell.

2.12.5 13.5 Grounding: how the data got there

The numbers on sheet 1 come from a single Python driver `work/timing_char_sweep.py` that:

1. Reads the nine `rtl/fub/*.sv` primitives.
2. Synthesizes each at two probe sizes (small + large) per corner through Yosys + ABC against ASAP7 RVT, using `work/merge_lib_corner.py`'s merged Liberty.
3. Reads ABC's `stime -c` output for each run to get delay (ps) and logic-level count.
4. Derives per-level slope and Tflop via linear extrapolation, drops them in `work/timing_char_data.csv`.

Then `work/build_characterization_xlsx.py` turns the CSV plus hand-walked FUB tables into the three-sheet workbook.

A parallel driver `work/abc_sweep.py` runs every STREAM library row (11 distinct blocks) through the same Yosys + ABC flow at TT / FF / SS x three target frequencies (99 cells, all clean, ~9 minutes wall-clock). Output is `work/timings.csv`; it's the "ground truth" measurement to calibrate the SWAGs on sheet 2 against – the SWAG model is first-order and underestimates real wide-bus buffering by ~10x, so production planning should treat the spreadsheet slack as an upper bound and lean on the CSV for the hard numbers.

2.12.6 13.6 The point

The same exercise can run against any ASIC technology you point the flow at: rerun the per-primitive probes against a different Liberty, regenerate the workbook, and the methodology recomputes. The characterization runs **once per (technology, corner)**; the rest of the workflow is a spreadsheet.

> The same workflow runs on an FPGA target – different delay model, different primitives, different gate units – and the worksheet shape stays the same; only the library columns change. That companion is the subject of a separate addendum README, not this document.

2.13 14. Navigation

- AI assistance guide: `CLAUDE.md`
- Product requirements: `PRD.md`
- Task tracking: `vault/Tasks/projects/asic-trials/timing_characterization/`
- Per-FUB synthesis recipes: `rtl/syn/SYNTHESIS_GUIDE.md`; branded PDF `docs/Timing_Characterization_Synthesis_Guide_v1.0.pdf` via `docs/generate_synthesis_guide_pdf.sh`
- Installing and running the open-source ASAP7 flow: handbook note `vault/handbook/asic/rtl-to-asap7-oss-flow.md`
- Measured cross-technology results (Artix-7 and Cyclone V frequency sweeps, carry-width / NAND-depth / multiplier-type parameter sweeps): `docs/baseline_results.md`, regenerated by `fpga/tools/baseline_report.py`
- Multi-flow constraint file: `rtl/syn/char_top.sdc`
- Constraint file: `rtl/syn/char_top.sdc`
- Open-source flow driver: `rtl/syn/Makefile`

- Design-time budget worksheet: `work/asap7_characterization.xlsx`
- Workbook generator: `work/build_characterization_xlsx.py`
- Per-primitive ASAP7 probe driver: `work/timing_char_sweep.py`
- Per-block STREAM library sweep: `work/abc_sweep.py`
- Merged ASAP7 RVT Liberty builder: `work/merge_lib_corner.py`
- Primitive characterization data: `work/timing_char_data.csv`
- STREAM library sweep results: `work/timings.csv`
- Components area: `projects/components/README.md`
- Repository guide: `/CLAUDE.md`